

Linking survey data to other data (03-05)

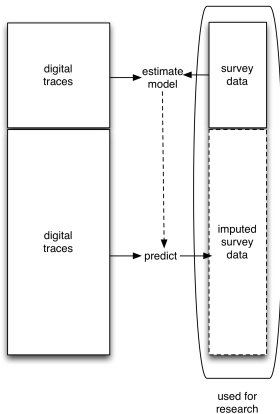
Matthew J. Salganik
Department of Sociology
Princeton University

Soc 596: Computational Social Science
Fall 2016

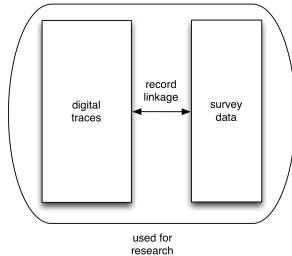


	Sampling	Interviews	Data environment
1st era	Area probability	Face-to-face	Stand-alone
2nd era	Random digital dial probability	Telephone	Stand-alone
3rd era	Non-probability	Computer-administered	Linked

Amplified asking



Enriched asking

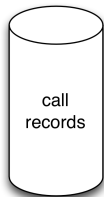


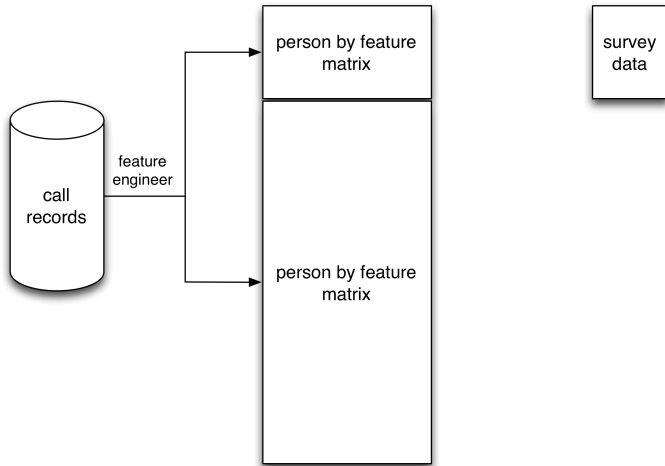
Note the different role of the big data in each case

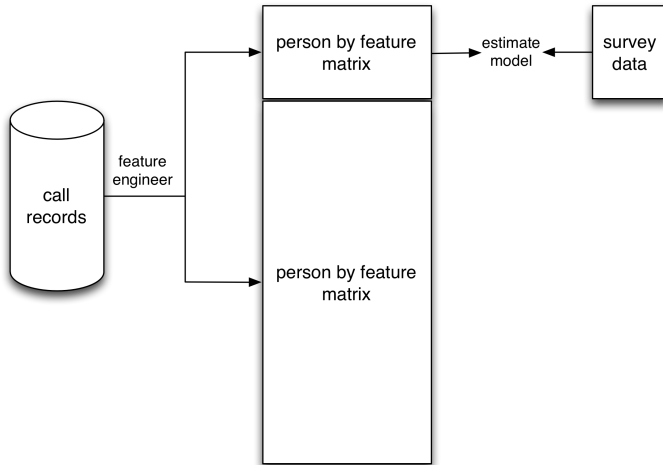
Predicting poverty and wealth from mobile phone metadata

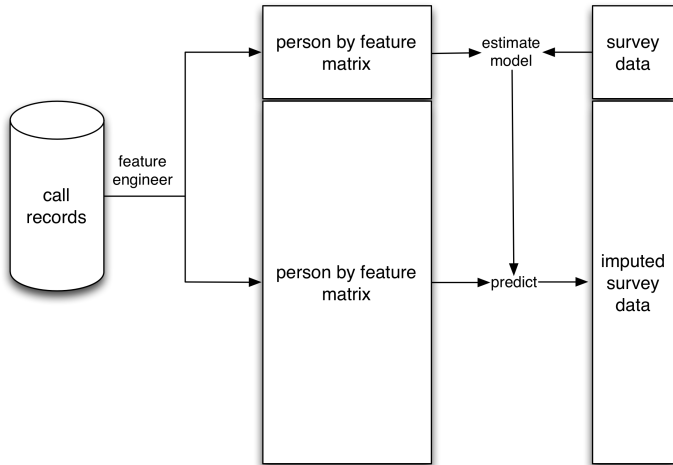
Joshua Blumenstock,^{1*} Gabriel Cadamuro,² Robert On³

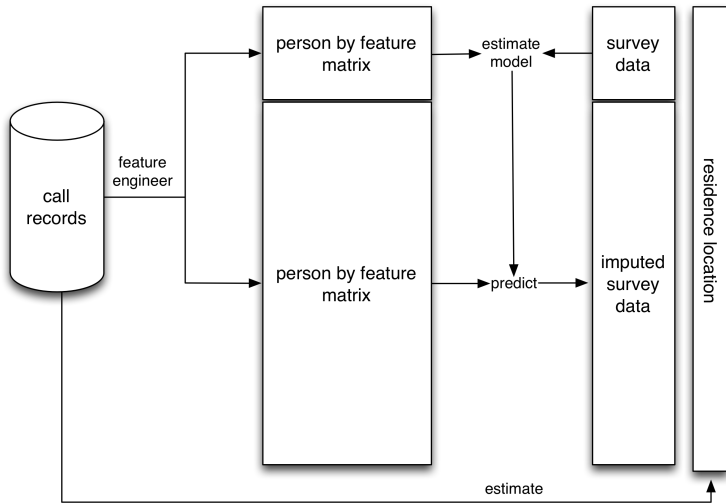


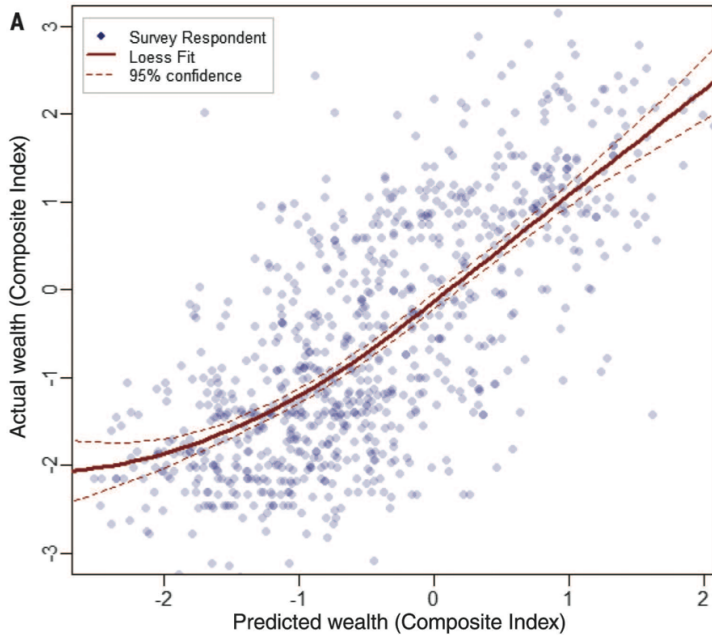


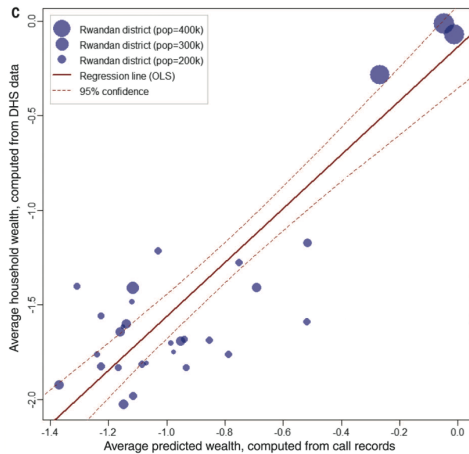


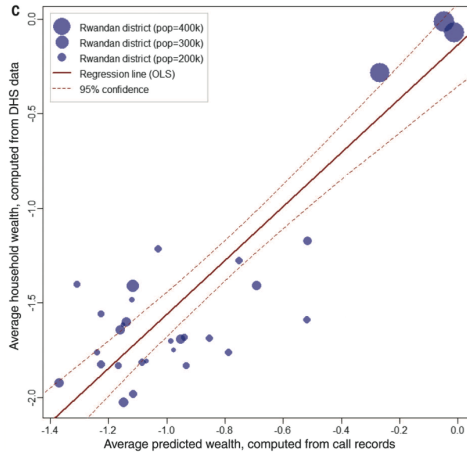












- ▶ 10 times faster
- ▶ 50 times cheaper

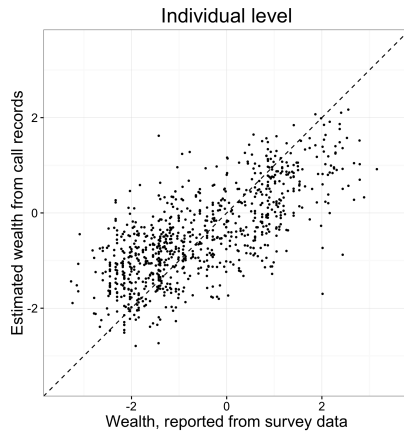


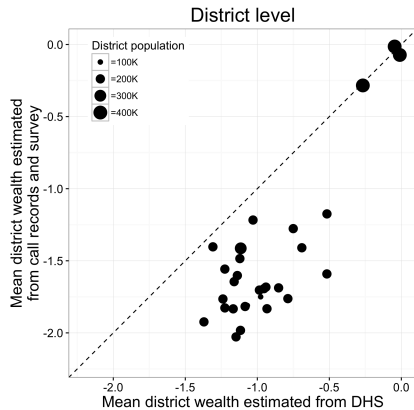
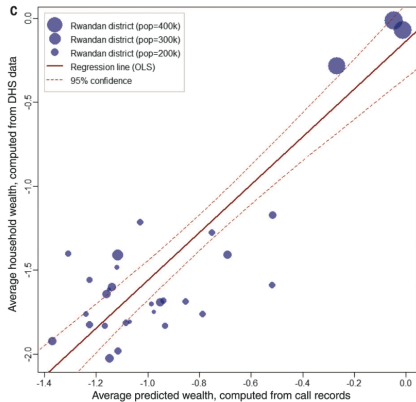
Readymades

+



Custommades





Predicting poverty and wealth from mobile phone metadata

Joshua Blumenstock,^{1*} Gabriel Cadamuro,² Robert On³

Calling for Better Measurement:

Estimating an Individual's Wealth and Well-Being
from Mobile Phone Transaction Records

Joshua E. Blumenstock
University of Washington
Seattle, WA
joshblum@uw.edu

Behavioral Modeling for Churn Prediction:

Early Indicators and Accurate Predictors of Custom Defection and Loyalty

Muhammad Raza Khan¹, Joshua Mano², Anikate Singh³, Joshua Blumenstock⁴
¹Information School, University of Washington, Seattle, WA, USA
Email: ¹mraz@uw.edu, ²joshuam@uw.edu, ³anikasingh@uw.edu, ⁴joshblum@uw.edu

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- ▶ better feature engineering
- ▶ new outcome variable
- ▶ out-of-sample test

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- ▶ outcome in data vs collected in a survey

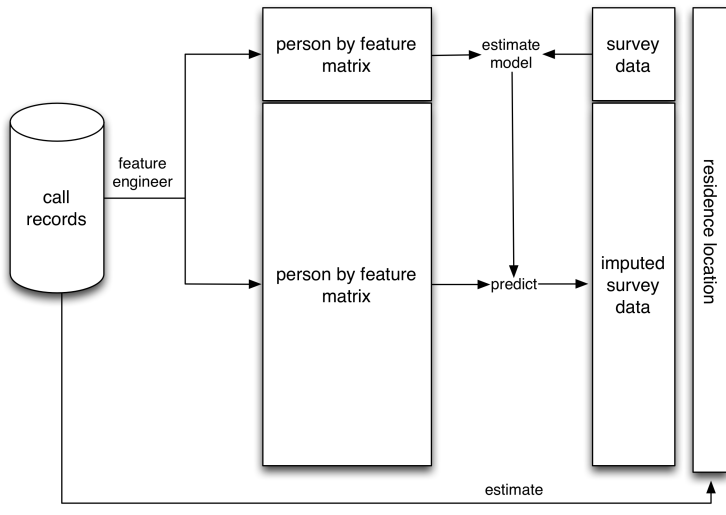
Combining satellite imagery and machine learning to predict poverty

Neal Jean,^{1,2*} Marshall Burke,^{3,4,5*}† Michael Xie,¹ W. Matthew Davis,⁴
David B. Lobell,^{3,4} Stefano Ermon¹

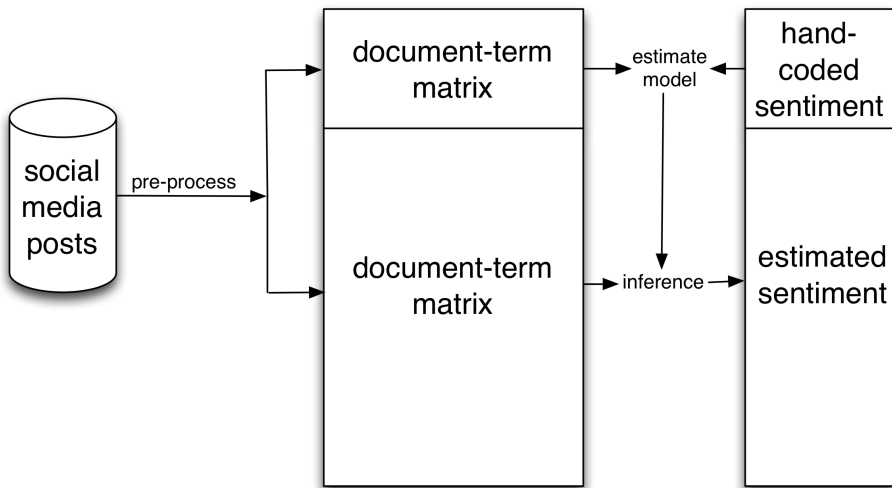
This paper is amazing and surprising. First a digression

Supervised learning:

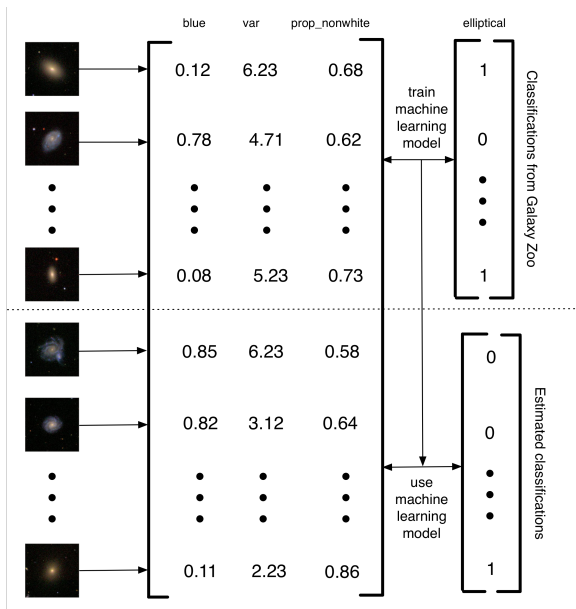
Lots of input-output pairs; goal is to develop a function that will predict the output from the input



See Chapter 3 of Salganik (2016)



See Chapter 2 of Salganik (2016)



See Chapter 5 of Salganik (2016)

Supervised learning:

Lots of input-output pairs; goal is to develop a function that will predict the output from the input

What if rather than engineering the features you could “learn” them automatically?

Deep learning

Yann LeCun^{1,2}, Yoshua Bengio³ & Geoffrey Hinton^{4,5}

<http://dx.doi.org/10.1038/nature14539>

“Have you tried deep learning?”

Deep learning + poverty + space

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Live demo:

<https://www.google.com/maps/place/Kigali,+Rwanda/@-1.9546259,30.0345059,26517m/data=!3m2!1e3!4b1!4m5!3m4!1s0x19dca4258ed8e797:0xf32b36a5411d0bc8!8m2!3d-1.9705786!4d30.1044288>

But, most people had been using night lights



https://www.nasa.gov/multimedia/imagegallery/image_feature_2480.html

Nightlights + survey data to estimate wealth in places without surveys

Jean et al. (2016):

Day pictures + Nightlights + survey data to estimate wealth in places without surveys

Predicting poverty

Satellite images can be used to estimate wealth in remote regions.

Neural network learns features in satellite images that correlate with economic activity

Daytime satellite photos capture details of the landscape



Convolutional Neural Network (CNN) associates features from daytime photos with nighttime intensity



Satellite nightlights are a proxy for economic activity

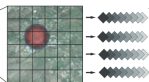


Daytime satellite images can be used to predict regional wealth

Household survey locations



CNN processes satellite photos of each survey site



Features from multiple photos are averaged



Ridge regression model reconstructs ground truth estimates of poverty

- Start with CNN pretrained on ImageNet (e.g. hamsters and weasels)

Predicting poverty

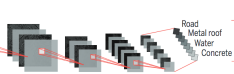
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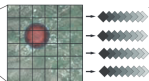


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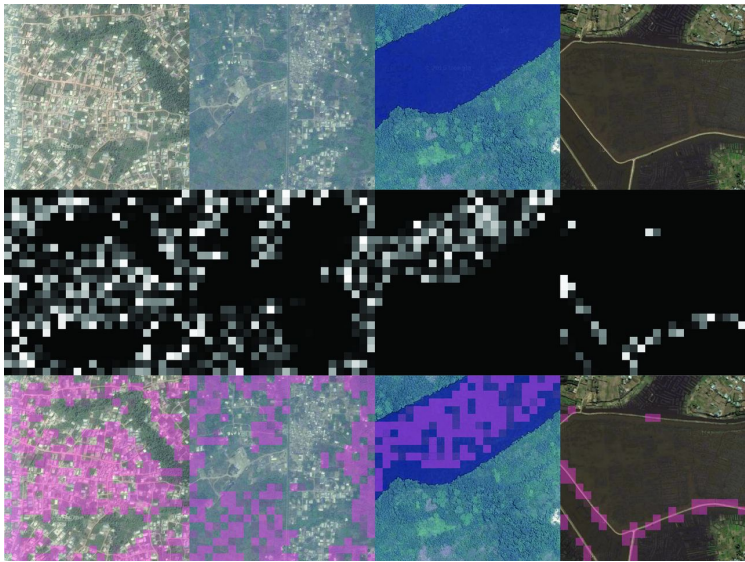


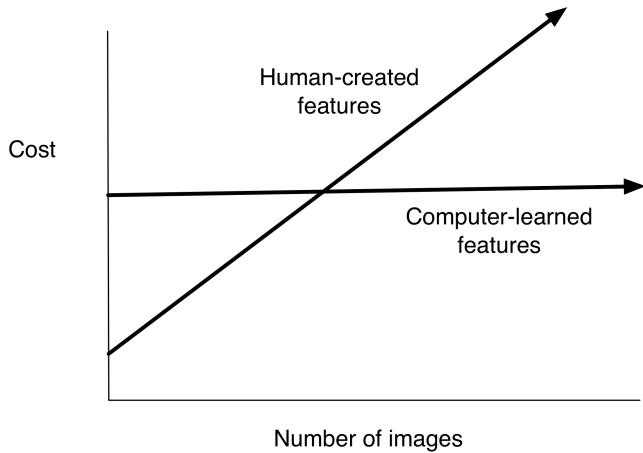
Features from multiple photos are averaged



Ridge regression model reconstructs ground truth estimates of poverty

- ▶ Start with CNN pretrained on ImageNet (e.g. hamsters and weasels)
- ▶ Train CNN to predict nightlights from day pictures (lots of training data)





Predicting poverty

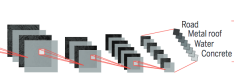
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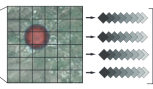


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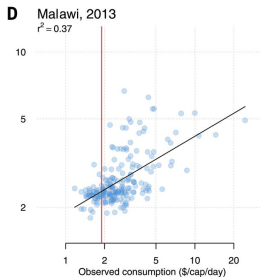
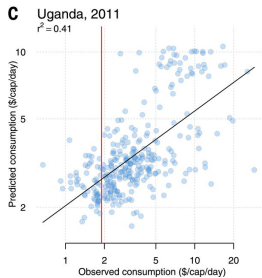
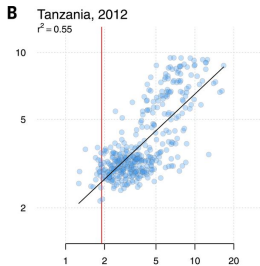
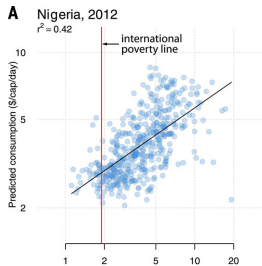


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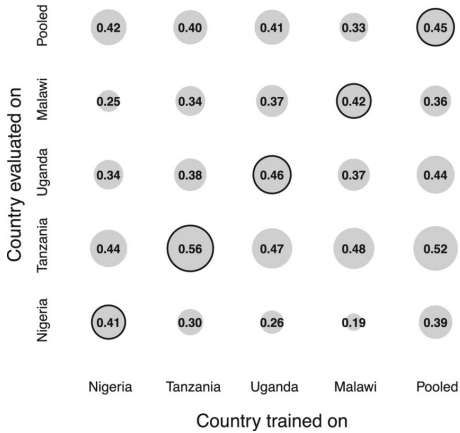
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- ▶ Start with CNN pretrained on ImageNet (e.g. hampsters and weasels)
- ▶ Train CNN to predict nightlights from day pictures (lots of training data)
- ▶ Take features from CNN and train ridge regression to predict cluster mean survey response

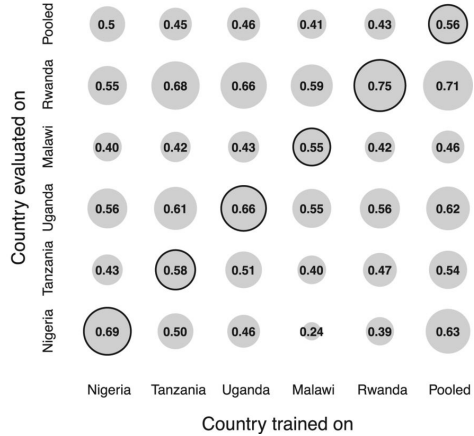


How could this figure be improved?

A Consumption expenditures



B Assets



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Readymade + Custommade

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Readymade + Readymade + (Quasi)
Custommade

Both Blumenstock et al (2015) and Jean et al (2016) estimate poverty in Africa?
Compare the strengths and weaknesses of the approaches.

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Uses Readymade linked to
Researcher-Custommade

Detecting influenza epidemics using search engine query data

Jeremy Ginsberg¹, Matthew H. Mohebbi¹, Rajan S. Patel¹, Lynnette Brammer², Mark S. Smolinski¹ & Larry Brilliant¹

Uses Readymade linked to
Researcher-Custommade

Is there a role for individual researchers collecting data in the age of the Readymades and Quasi Custommades?

How should governments collect large general purpose datasets if they will be combined with Readymades?